SAI KRISHNA

AI/ML ENGINEER | GENERATIVE AI, RAG, AGENTIC AI & MLOPS

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PROFESSIONAL SUMMARY

- **AI/ML Engineer** with **4+ years** building production AI across the **full lifecycle**, from requirements and architecture to deployment, monitoring, and iteration, spanning **Generative AI**, **agentic AI**, **fraud detection**, and **computer vision**.
- Build **RAG pipelines** and **LangGraph** agentic workflows with **LangChain**, **Hugging Face**, and **Pinecone**, grounding LLM answers in source documents with query decomposition, conditional routing, and **human-in-the-loop** validation.
- Design scalable ML systems end to end: **PySpark** and **Kafka** pipelines processing **1M+ transactions daily**, **XGBoost** risk-scoring models, and **FastAPI** microservices serving **10K+ queries/day** at low latency.
- Make AI measurable and reliable with **LLM evaluation pipelines** (faithfulness, hallucination risk, regression testing), **MLflow** tracking, production monitoring, and **CI/CD** on **Docker** and **Kubernetes** across **AWS** and **Azure**.
- Partner with product, risk, and engineering stakeholders in **enterprise** and **regulated finance** settings to turn requirements into AI products, backed by clear architecture documentation, evaluation reports, and runbooks.

TECHNICAL SKILLS

Languages & Databases	Python, SQL, Java, JavaScript, TypeScript, PostgreSQL, MySQL, SQL Server, DynamoDB, Redis, OpenSearch, Vector Databases
ML & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, Pandas, NumPy, SciPy, Matplotlib, Seaborn, QLoRA, PEFT, CNN, ResNet, VGG16, Transfer Learning, Data Augmentation, OpenCV
LLMs & Prompting	Amazon Bedrock (Claude Sonnet/Opus), GPT-4/5, LLaMA 2/3/4, Gemini, FinGPT, Prompt Engineering, CoT Reasoning, Few-Shot, ReAct Prompting, Structured JSON Outputs
Generative AI & RAG	RAG, GraphRAG, Agentic AI, Multi-Agent Systems, Semantic Chunking, Vector Embeddings, Hybrid Search, Cross-Encoder Reranking, Query Decomposition, Human-in-the-Loop, AgentCore
LLM Frameworks & Tools	LangChain, LangGraph, CrewAI, Pinecone, FAISS, Chroma DB, OpenSearch Serverless, Azure AI Search, Bedrock Knowledge Bases, Bedrock Guardrails
LLM Evaluation & Safety	Answer Faithfulness, Hallucination Detection, Retrieval Relevance, LLM-as-Judge, RAGAS, LangSmith, Regression Testing, Red-Teaming, BLEU, ROUGE
MLOps & LLMOps	MLflow, Docker, Kubernetes, Terraform IaC, GitHub Actions CI/CD, Model Versioning, Canary Deployments, Drift Monitoring, Structured Logging, Datadog, CloudWatch, Sentry
Cloud Platforms	AWS (SageMaker, Bedrock, Lambda, Step Functions, EKS, ECS, ECR, Glue, Athena, EC2, API Gateway), Azure (ML Studio, AI Search, Azure OpenAI, AKS), GCP (Vertex AI)
APIs & Data Pipelines	FastAPI, REST APIs, Async APIs, Microservices, MCP, A2A, OpenAI APIs, Apache Spark, PySpark, Kafka, Airflow, Feature Engineering, Data Quality Validation
Model Evaluation	Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, Threshold Tuning, Class Imbalance (SMOTE), SHAP, Cross-Validation, Statistical Testing
Tools & Practices	Git, GitHub, GitLab, Pytest, OOP, Design Patterns, Agile/Scrum, Jira, React.js, Claude Code, Codex, Kiro

PROFESSIONAL EXPERIENCE

AI/ML Engineer | **Adobe**, San Francisco, CA, USA ***Jan 2025 – Present***

- Partnered with product managers, engineering teams, and business stakeholders to turn enterprise requirements into scalable **Generative AI** solutions for document intelligence, semantic search, and knowledge retrieval.
- Engineered end-to-end **RAG pipelines** with **LangChain**, **Hugging Face**, and **Pinecone** (document ingestion, chunking, embeddings, vector indexing), improving retrieval relevance **11%** and cutting document processing time **25%**.
- Developed **LangGraph** agentic workflows with query decomposition, conditional routing, state management, and **human-in-the-loop** validation, reducing unsupported LLM responses **10%** through retrieval grounding and response validation.
- Designed and optimized **FastAPI** microservices and **RESTful APIs** for enterprise AI integration with asynchronous inference, request validation, and exception handling, reducing response latency **20%** while serving **10K+ queries per day**.
- Created systematic **LLM evaluation pipelines** scoring retrieval relevance, answer faithfulness, **hallucination risk**, and response consistency using benchmark datasets, prompt optimization, and **regression testing**.
- Deployed containerized AI services with **Docker** and **Kubernetes** on **AWS** and **Azure**, automating tests, image builds, and release validation in **GitHub Actions CI/CD**, cutting deployment time **30%**.
- Implemented **MLflow** experiment tracking, model versioning, structured logging, latency monitoring, and production alerting, contributing to a **20% reduction** in recurring production incidents.
- Produced technical architecture documentation, evaluation reports, deployment runbooks, and operational handover materials to support production reliability, cross-functional collaboration, and long-term maintainability.

Tech Stack: *Python, LangChain, LangGraph, Hugging Face, Pinecone, FastAPI, Docker, Kubernetes, AWS, Azure, GitHub Actions, MLflow*

AI/ML Research Assistant | **University of North Texas**, Denton, TX, USA ***Jan 2024 – Dec 2024***

- Conducted applied deep learning and **computer vision** research for **medical image classification** using **TensorFlow**, **Keras**, and **PyTorch**, building reproducible training and experimentation workflows.
- Implemented and benchmarked **CNN**, **ResNet**, and **VGG** architectures, applying **transfer learning**, image preprocessing, **data augmentation**, and hyperparameter tuning to improve model generalization.
- Evaluated models with accuracy, precision, recall, **F1-score**, and **confusion matrices**, running error analysis to surface misclassification patterns and model limitations.
- Collaborated with faculty researchers to interpret findings, document methodology, and communicate model performance, reproducibility, and technical trade-offs.

Tech Stack: *Python, TensorFlow, Keras, PyTorch, CNN, ResNet, VGG, Transfer Learning, Data Augmentation*

Machine Learning Engineer | **Wells Fargo**, Hyderabad, India ***Jan 2022 – Dec 2023***

- Partnered with fraud analytics, risk management, and engineering stakeholders to define ML requirements and develop scalable **transaction fraud detection** and **risk-scoring** solutions.
- Built distributed data pipelines with **Python**, **SQL**, **PySpark**, and **Kafka** processing **1M+ transactions daily**, cutting data processing time **30%** through partitioning and query optimization.
- Engineered behavioral transaction features and trained **XGBoost** and **Scikit-learn** fraud classifiers with feature selection, hyperparameter tuning, and **class-imbalance handling**, improving F1-score **7%** and fraud recall **25%**.
- Optimized decision thresholds on the precision-recall trade-off, reducing **false-positive alerts 18%** while supporting risk-based transaction investigations.
- Developed **FastAPI**-based RESTful inference services and deployed containerized ML applications with **Docker** and **Kubernetes** on **AWS**, reducing inference latency **23%** through runtime optimization.
- Automated **CI/CD** workflows covering Git, unit testing, model validation, container builds, and Kubernetes deployments, reducing deployment time **35%** and improving release consistency.
- Established **MLflow** experiment tracking, model versioning, data-quality validation, and production monitoring, contributing to a **20% reduction** in recurring production incidents.
- Documented model architecture, evaluation results, deployment procedures, and operational runbooks, supporting production handover, troubleshooting, and collaboration with fraud operations teams.

Tech Stack: *Python, SQL, PySpark, Kafka, Scikit-learn, XGBoost, FastAPI, Docker, Kubernetes, AWS, MLflow, Git*

PROJECTS

Enterprise RAG-Based Knowledge Assistant ***In Progress***

- Building a **Retrieval-Augmented Generation (RAG)** application for knowledge-base question answering, integrating document ingestion, **semantic chunking**, vector embeddings, and **Pinecone** similarity search.
- Delivering **citation-backed** LLM responses through **FastAPI** RESTful APIs, with retrieval evaluation and **Docker** packaging.

Tech Stack: *Python, LangChain, Hugging Face, Pinecone, FastAPI, Docker*

Agentic AI Workflow Automation System ***In Progress***

- Designing a **multi-agent** workflow automation system in **LangGraph** with specialized planning, retrieval, execution, and validation agents coordinated through **conditional routing** and shared state.
- Implementing **human approval checkpoints** and API-based orchestration with **FastAPI**, targeting deployment on **AWS** with **Docker**.

Tech Stack: *Python, LangGraph, LangChain, FastAPI, AWS, Docker*

EDUCATION

Master of Science (M.S.), Business Analytics | University of North Texas | CGPA: 3.75/4.0

CERTIFICATIONS

Microsoft Certified: Azure Data Fundamentals (DP-900) | Microsoft Certified: Power BI Data Analyst Associate (PL-300)
AWS Certified Cloud Practitioner | Anthropic: Claude 101 | Anthropic: AI Fluency: Framework & Foundations